4.1 Homeostasis and extrinsic motivation

Several computational implementations of emotions involve homeostatic variables, drives and motivations. The notion of internal drives originates from the Drive Reduction Theory developed by Hull (1943), which identifies drive reduction as a central cause of learning. These innate drives are also known as primary reinforcers, as their rewarding nature is hard-wired in our system (due to evolutionary benefit). An example of a homeostatic variable is energy/sugar level, which has a temporal dynamic, an associated drive when in deficit (hunger) and can be satiated by an external influence (food intake). The reader might now question why machines even need something like 'hunger'. However, for a robot the current energy level shows similarity to human sugar levels (and body integrity and pain show similarity to a robot's mechanical integrity, etc.). Thereby, homeostasis is a useful concept to study in machines as well (see also the remark about bio-inspiration in the Introduction). There is a vast literature on motivated reinforcement learning, see e.g. Konidaris and Barto (2006) and Cos et al (2013), mainly for its potential to naturally switch between goals. Early implementations of these ideas outside the reinforcement learning framework were by Cañamero (1997a,b).

We denote a homeostatic variable by h_t, where t identifies the dependency of this variable on time. The organism's full physiological state is captured by $H_t = \{h_{1,t}, h_{2,t}..h_{N,t}\}$, where $h_{i,t}$ indicates the ith homeostatic variable. Each homeostatic variable has a certain set point $H^\star = \{h_1^\star, h_2^\star..h_N^\star\}$ (Keramati and Gutkin, 2011). Furthermore, each homeostatic variable is affected by a set of external resources, associated to a particular action or state. For example, a particular homeostatic variable may increase upon resource consumption, and slightly decrease with every other action (Konidaris and Barto, 2006). More formally, denoting resource consumption by $\bar{a}$ and the presence of a resource by $\bar{s}$, a simple homeostatic dynamic would be

$$h_{i,t+1} = \begin{cases} h_{i,t} + \psi(s_t, a_t) & \text{if } a_t \in \bar{a}, s_t \in \bar{s} \\ h_{i,t} - \epsilon & \text{otherwise} \end{cases} \tag{5}$$

for a resource effect of size $\psi(s_t, a_t)$. We can also explicitly identify a *drive* as the difference between the current value and setpoint, i.e. $d_{i,t} = |h_i^\star - h_{i,t}|$ (Cos et al, 2013). The overall drive of the system can then be specified by

$$D_t = \sum_{i=1}^{N} \theta_i d_{i,t} = \sum_{i=1}^{N} \theta_i |h_i^\star - h_{i,t}| \tag{6}$$

where we introduced θ_i to specify the weight or importance of the i-th homeostatic variable. Most examples take the absolute difference between current value and setpoint (i.e. the L_1 norm) as shown above. However, we can consider the space of homeostatic variables $H \in \mathbb{R}^N$ and in principle define any distance function in this space with respect to the reference point $H^\star$ (see e.g. Figure 2 for a Euclidean distance example).

The weight of each homeostatic variable (θ_i) does not need to be fixed in time. For example, Konidaris makes it a non-linear function of the current homeostatic level $h_{i,t}$ and a priority parameter $\rho_{i,t}$: $\theta_{i,t} = f(h_{i,t}, \rho_{i,t})$. The former dependence

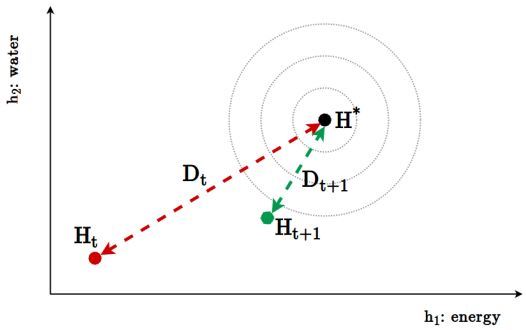

Fig. 2 Schematic illustration of homeostasis and drives. The figure shows a two-dimensional homeostatic space consisting (as an example) of energy (h_1) and water level (h_2). The set point ($H^\star$) indicates the desired values for the homeostatic variables. At the current timepoint t the agent's homeostatic status is H_t (red). The associated drive D_t can be visualize as the distance to the set point. Note that we use the Euclidean distance for the drive here (i.e. $D_t = ||H^\star - H_t||_2$), while the text describes the L_1-norm example (i.e. $D_t = ||H^\star - H_t||_1$, equation 6). We are free to choose any distance metric in homeostatic space. After taking an action the new homeostatic status becomes H_{t+1} (green), in this case bringing both homeostatic levels closer to their set point. The difference between the drives at both timepoints has been associated to reward and joy (see Section 6.1). Figure partially based on Keramati and Gutkin (2011).

allows priorities (i.e. rewards) to scale non-linearly with the sensory input levels (an idea reminiscent of Prospect Theory (Kahneman and Tversky, 1979)). The priority parameters $\rho_{i,t}$ can be estimated online, for example assigning more importance to resources which are harder to obtain (i.e. that should get priority earlier). As a final note on homeostatic RL systems, note that internal variables need to be part of the state-space as well. One can either include all homeostatic variables and learn generic Q-values, or include only the dominant drive and learn drive-specific Q-values (Konidaris and Barto, 2006).

The connection between drives/homeostasis and emotions is partially reflected in Damasio's somatic marker hypothesis (Damasio, 1994), stating that emotions are the result of bodily sensations. In general, we identify two ways in which homeostatic systems have been used to elicit emotions. The first elicits categorical emotions from a subset of homeostatic variables, while the second derives an overall well-being W from the sum of the homeostatic dimensions.

One of the first RL systems deriving emotions from homeostasis was by Gadanho and Hallam (1998, 2001). They describe an extensive set of internal variables (drives), including e.g. hunger (rises per timestep in lack of resources), pain (rises with collisions), restlessness (rises with non-progress) and temperature (rises with high motor usage). Emotions are related to these physiological variables, e.g. happiness is derived from the frequent motor use or decreasing hunger, sadness from low energy, fear from collisions (with less sensitivity if the agent is hungry or restless), and anger from high restlessness. Similar ideas are put forward by Coutinho et al (2005), who specifies a more biological homeostasis: blood sugar (increases with food intake), endorphine (increases with play), energy (increases with bed rest), vascular volume (increases with water intake) and body integrity (decreases

Table 2 Overview of most frequently investigated homeostatic dimensions, their associated drive in case of deficit, and the papers in which example implementations can be found.

Homeostasic variable	**Drive**	**Papers**
Food/energy	Hunger	(Gadanho and Hallam, 2001), (Salichs and Malfaz, 2012), (Coutinho et al, 2005) (Von Haugwitz et al, 2012) (Goerke, 2006) (Tanaka et al, 2004)
Water level	Thirst	(Salichs and Malfaz, 2012) (Coutinho et al, 2005)
Body integrity	Pain	(Gadanho and Hallam, 2001) (Coutinho et al, 2005) (Tanaka et al, 2004) (Lee-Johnson et al, 2010)
Activity	Restlessness	(Gadanho and Hallam, 2001) (Coutinho et al, 2005) (Von Haugwitz et al, 2012)
Energy (movement)	Sleep/tiredness	(Salichs and Malfaz, 2012) (Coutinho et al, 2005) (Von Haugwitz et al, 2012) (Goerke, 2006) (Tanaka et al, 2004)
Social interaction	Loneliness	(Salichs and Malfaz, 2012)

with obstacle collision). Similar examples of homeostatic emotions can be found in Von Haugwitz et al (2012), Tanaka et al (2004) and Goerke (2006).

A second group of implementations first defines the overall *well-being* (W). An example of a well-being specification is

$$W_t = K - D_t = K - \sum_{i=1}^{N} \theta_i |h_i^\star - h_{i,t}| \tag{7}$$

where K denotes a reference value. Compared to the previous paragraph, now *all* internal variables (instead of subsets) are combined into a single emotion. Some papers leave the specification of well-being as their emotion (Gadanho, 2003). Others actually identify the positive or negative difference in well-being as happy and unhappy (Salichs and Malfaz, 2012) or 'hedonic value' (Cos et al, 2013).

In conclusion, there have been numerous approaches to homeostatic systems in emotional implementations. A summary of some of the most frequently encountered homeostatic dimensions is shown in Table 2. Although most papers use slightly different specifications for their homeostatic dimensions, it is usually a matter of labelling that does not affect the underlying principle. Homeostatic variables provide a good way to naturally implement goal and task switching. The implementation of this functionality usually involves reward modification, which is covered in Section 6.1.

4.2 Appraisal and intrinsic motivation

Appraisal theory is an influential psychological emotion theory (see Section 2). Appraisals are domain independent elements that provide (affective) meaning to a particular stimulus. As such, they are a basis for emotion elicitation, as different combinations of appraisal dimensions have different associated emotions. Examples of appraisal dimensions are novelty, recency, control and motivational relevance. These terms of course refer to abstract cognitive concepts, but in RL literature